

Local Langlands conjecture for the double covers of GL_2 and GL_3

Explicit parameters, central characters, and a term-by-term proof

Research note prepared for Rui Chen

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Abstract

For the standard Kazhdan–Patterson double covers of $\mathrm{GL}_2(F)$ and $\mathrm{GL}_3(F)$, where F is any finite extension of $\mathbb{Q}_p$, we construct the finite-to-one map requested in Conjecture 0.0.3 of the supplied note. We compute the two extensions defining the L -group, remove the dependence on the incarnation parameter η , and give exhaustive tables of genuine irreducible representations and their parameters. The proof uses the established discrete-series metaplectic correspondence, the classification of genuine tempered representations, and the Langlands classification. The intermediate Weil–Deligne assignment is bijective onto an explicitly specified subset of parameters. Its infinitesimal Weil parameter has fibers of size at most two in rank two and at most four in rank three. These bounds are attained. The argument includes residue characteristic two.

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1 Statement, scope, and conventions

Write $G_r = \mathrm{GL}_r(F)$, $r = 2, 3$, and fix the nontrivial character $\epsilon : \mu_2 \hookrightarrow \mathbb{C}^\times$. All representations are smooth, complex, irreducible, admissible, and ϵ -genuine unless stated otherwise. We use the standard untwisted Kazhdan–Patterson incarnation, namely $c = 0$ in Gan–Gao [4, §16.2]. Thus, with $Y = \mathbb{Z}^r$ and its standard basis e_i ,

$$Q(y) = \sum_{i < j} y_i y_j, \quad D(y, z) = \sum_{i < j} y_i z_j, \quad Q(e_i - e_j) = -1. \quad (1)$$

The resulting degree-two cover with incarnation parameter $\eta : Y^{\mathrm{sc}} \rightarrow F^\times$ is denoted $\overline{G}_{r,\eta}$. Here $Y^{\mathrm{sc}} = \{y : \sum_i y_i = 0\}$. The parameter η is part of a presentation, not an additional isomorphism class of covers for GL_r ; an explicit transport is given in §8.

Let $\nu(a) = |a|_F$ and, on G_m , also write $\nu = |\det|_F$. Fix reciprocity $\mathrm{rec}_F : W_F^{\mathrm{ab}} \simeq F^\times$ with geometric Frobenius mapping to a uniformizer. We identify characters of $F^\times$ with characters of W_F and put $\nu(w) = |\mathrm{rec}_F(w)|_F$. Define

$$X^+(F) = \{\lambda : F^\times \rightarrow \mathbb{C}^\times \text{ smooth} : \lambda(-1) = 1\}.$$

For an irreducible Weil representation ρ , the notation $\det \rho(-1)$ means evaluation at -1 of the character corresponding to $\det \rho$ by reciprocity; it does not require choosing an element “ -1 ” in W_F .

We use $S_m = \mathrm{Sym}^{m-1}(\mathbb{C}^2)$ for the m -dimensional representation of $\mathrm{SL}_2(\mathbb{C})$. A Frobenius-semisimple Weil–Deligne parameter is expressed as

$$\Phi = \bigoplus_j \rho_j \boxtimes S_{m_j}, \quad \sum_j m_j \dim \rho_j = r. \quad (2)$$

The ρ_j are irreducible smooth Weil representations, possibly unramified twists of unitary ones. Set

$$R_\Phi(w) = \Phi\left(w, \mathrm{diag}(\nu(w)^{1/2}, \nu(w)^{-1/2})\right). \quad (3)$$

This is the Weil representation in the usual (R, N) description. For a block it is

$$R_{\rho \boxtimes S_m} = \bigoplus_{j=0}^{m-1} \rho \nu^{(m-1)/2-j}. \quad (4)$$

Thus forgetting N means (3), not restriction to $W_F \times \{1\}$. Both operations produce Weil parameters, but they are different maps.

Definition 1.1. A block $\rho \boxtimes S_m$ is called allowed in this note if

$$(\det \rho(-1))^m = 1. \quad (5)$$

Let $\mathcal{P}_r^+$ be the parameters (2) all of whose blocks are allowed, modulo equivalence. For rank three let

$$\tilde{\mathcal{P}}_3^+ = \{(\Phi, \kappa) : \Phi \in \mathcal{P}_3^+, \kappa : F^\times \rightarrow \mathbb{C}^\times, \kappa^2 = \det \Phi\} / \simeq.$$

Only Φ is conjugated; the character κ is retained.

Theorem 1.2. For $r = 2, 3$ and every η , Conjecture 0.0.3 of [1] holds for $\overline{G}_{r,\eta}$. More explicitly, after the normalization and transport specified below, there are bijections

$$\mathrm{Irr}_\epsilon(\overline{G}_{2,\eta}) \longleftrightarrow \mathcal{P}_2^+, \quad \mathrm{Irr}_\epsilon(\overline{G}_{3,\eta}) \longleftrightarrow \tilde{\mathcal{P}}_3^+.$$

Composing with (3) and the chosen splitting of the L -group gives

$$\mathcal{L}_\eta : \mathrm{Irr}_\epsilon(\overline{G}_{r,\eta}) \longrightarrow \{\text{splittings } W_F \rightarrow {}^L \overline{G}_{r,\eta}\} / \overline{G}_r^\vee.$$

Its fibers have size at most 2 for $r = 2$, and at most 4 for $r = 3$.

The theorem concerns the finite-to-one assertion actually written in the supplied note. It does not assert all the additional desiderata of a full local Langlands correspondence. In particular, surjectivity onto every parameter is neither claimed nor true for this construction. The proof is a deduction from established classification and transfer theorems, not a new proof of those theorems. The supplied volume supplies the L -group formalism; the later references [9, 10] supply the representation-theoretic input in a form that handles all cases used here.

2 The lattices and the two L -group extensions

Put $A = F^\times / F^{\times 2}$ and $(a, b) = (a, b)_{F,2}$. For $S(y) = \sum_i y_i$, (1) gives

$$B_Q(y, e_j) = S(y) - y_j.$$

Since $Q(\alpha^\vee) = -1$, every modified coroot is $\beta_i = 2(e_i - e_{i+1})$ and

$$L_{\text{sc}} = Y_{Q,2}^{\text{sc}} = 2Y^{\text{sc}}.$$

The equations $B_Q(y, e_j) \in 2\mathbb{Z}$ give, directly,

$$\begin{array}{c|c|c} r & L = Y_{Q,2} & L/L_{\text{sc}} \\ \hline 2 & 2\mathbb{Z}^2 & \mathbb{Z} \quad (y \mapsto S(y)/2) \\ 3 & 2\mathbb{Z}^3 + \mathbb{Z}e_c & \mathbb{Z} \quad (y \mapsto S(y)), \quad e_c = e_1 + e_2 + e_3. \end{array} \quad (6)$$

In rank three, if $S(y) = 0$, the common parity of the three entries must be even, proving the indicated kernel. The map $S : L \rightarrow \mathbb{Z}$ is onto since $S(2e_1 - e_c) = -1$.

2.1 Dual groups, including the rank-three isogeny

The modified root datum gives

$$H_2 := \overline{G}_2^\vee = \text{GL}_2(\mathbb{C}), \quad H_3 := \overline{G}_3^\vee = \{(g, z) \in \text{GL}_3(\mathbb{C}) \times \mathbb{C}^\times : \det g = z^2\}. \quad (7)$$

In rank two, the characters $2e_1, 2e_2$ are the usual two diagonal weights. In rank three the characters $2e_i$ give the diagonal entries of g , while e_c gives z ; the relation $\sum_i 2e_i = 2e_c$ gives $\det g = z^2$. The roots are $2(e_i - e_j)$, as required. This verifies the particular instance of [4, §16.2, p. 267] directly.

Although H_3 is abstractly isomorphic to $\text{GL}_3(\mathbb{C})$, its projection to the first factor is a double isogeny, not that isomorphism. An explicit isomorphism and inverse are

$$(g, z) \mapsto z^{-1}g, \quad h \mapsto (h(\det h)^{-1}, (\det h)^{-1}). \quad (8)$$

Retaining the pair (g, z) is essential for compatibility with central characters and metaplectic transfer.

2.2 The second extension

Use exactly the presentation in the supplied note:

$$\mathcal{E}_{Q,2} = L \times A, \quad (y, u)(z, v) = (y + z, uv[(-1)^{D(y,z)}]).$$

For $\eta_i = \eta(e_i - e_{i+1})$ one has $\eta(\beta_i) = \eta_i^2$, hence

$$s_\eta(\beta_i) = (\beta_i, 1).$$

Thus $\mathcal{Q}_\eta = \mathcal{E}_{Q,2}/s_\eta(L_{sc})$ and its dual $E_{2,\eta}$ are independent of η in these coordinates.

For $r = 2$, $D(y, z)$ is divisible by four on L , so

$$\mathcal{Q}_\eta \simeq \mathbb{Z} \times A, \quad E_{2,\eta} \simeq \mathbb{C}^\times \times A.$$

For $r = 3$, write $m = S(y)$, $n = S(z)$. The common-parity condition gives

$$D(y, z) \equiv mn \pmod{2}, \quad Q(y) \equiv m \pmod{2}.$$

Consequently

$$\mathcal{Q}_\eta \simeq \mathbb{Z} \times A, \quad (m, u)(n, v) = (m + n, uv[-1]^{mn}). \quad (9)$$

The explicit change of coordinate

$$(m, u) \mapsto (m, u[-1]^{\binom{m}{2}})$$

turns (9) into the direct-product law, because $\binom{m+n}{2} = \binom{m}{2} + \binom{n}{2} + mn$. For $a \in A$ its character defining a splitting of $E_2 \rightarrow A$ is

$$t_a(m, u) = (u, a)(-1, a)^{\binom{m}{2}}. \quad (10)$$

This is multiplicative in both (m, u) and a , so it proves the splitting rather than merely counting extension classes.

2.3 The first twist and a distinguished splitting

The first twist is $c_1(a, b)(y) = (a, b)^{Q(y)}$. In rank two it is identically one. In rank three it is $(a, b)^{S(y)}$. Choose a nontrivial additive character ψ and its normalized Weil index γ_ψ with convention

$$\gamma_\psi(ab) = \gamma_\psi(a)\gamma_\psi(b)(a, b), \quad \gamma_\psi(a^2) = 1, \quad \gamma_\psi(a)^2 = (a, -1). \quad (11)$$

Then $a \mapsto \gamma_\psi(a)^{S(y)}$ splits E_1 in the rank-three central coordinate. Combining this with (10) splits the Baer sum. Pushout and pullback therefore yield

$${}^L\overline{G}_2 \simeq H_2 \times W_F, \quad {}^L\overline{G}_3 \simeq H_3 \times W_F. \quad (12)$$

Rank two needs no ψ for this normalization. In rank three the normalization can be characterized on the covering torus by

$$\chi_\psi(\mathbf{s}(e_i(a^2))) = 1, \quad \chi_\psi(\mathbf{s}(aI_3)) = \gamma_\psi(a)^{-3} = \gamma_\psi(a), \quad (13)$$

with $\chi_\psi|_{\mu_2} = \epsilon$. Here $\mathbf{s}$ is the standard torus section. These are [4, (16.8)] in this case; they also follow from its torus cocycle and (11). We use this distinguished splitting from now on.

Under (12), the full continuous semisimple Weil-parameter targets are all two-dimensional semisimple Weil representations in rank two, and all pairs (R, κ) with $\dim R = 3$ and $\det R = \kappa^2$ in rank three. The image restrictions in Definition 1.1 belong to the correspondence, not to the definition of the L -group.

3 The genuine torus and the central character

For $t = \text{diag}(t_1, \dots, t_r)$ and $u = \text{diag}(u_1, \dots, u_r)$ the standard torus cocycle is

$$\mathbf{s}(t)\mathbf{s}(u) = \prod_{i < j} (t_i, u_j) \mathbf{s}(tu). \quad (14)$$

The nondegeneracy of the Hilbert pairing shows that a diagonal t is central in $\overline{T}$ exactly when $\prod_{i \neq j} t_i$ is a square for every j . Therefore

$$p(Z(\overline{T}_2)) = \{\text{diag}(b_1^2, b_2^2)\}, \quad p(Z(\overline{T}_3)) = \{aI_3 \text{diag}(b_1^2, b_2^2, b_3^2)\}.$$

Here $p : \overline{G} \rightarrow G$ is the covering projection. The Stone–von Neumann theorem for the covering torus associates one genuine irreducible representation to each genuine character of its center; see [4, §8.2–8.3]. In both ranks its dimension is $|A|$: the commutator quotient has order $|A|^2$.

For rank two, write this representation as $\tau(\lambda_1, \lambda_2)$, where $\lambda_i \in X^+(F)$, with central character

$$\omega_\tau(\zeta \mathbf{s}(\text{diag}(b_1^2, b_2^2))) = \epsilon(\zeta) \lambda_1(b_1) \lambda_2(b_2). \quad (15)$$

For rank three, the analogous representation is $\tau_\psi(\lambda_1, \lambda_2, \lambda_3; \kappa)$, where

$$\lambda_i \in X^+(F), \quad \kappa^2 = \lambda_1 \lambda_2 \lambda_3, \quad \omega_\tau(\zeta \mathbf{s}(aI_3 \text{diag}(b_i^2))) = \epsilon(\zeta) \gamma_\psi(a) \kappa(a) \prod_i \lambda_i(b_i). \quad (16)$$

The last formula is independent of the presentation: replacing a by ac^2 and all b_i by b_i/c changes its value by $\kappa(c)^2 / \prod_i \lambda_i(c) = 1$. Changes of signs in the b_i are harmless precisely because $\lambda_i(-1) = 1$. Multiplicativity follows from (14) and (11).

The center of the entire cover is smaller in rank two:

$$Z(\overline{G}_2) = p^{-1}(F^{\times 2} I_2), \quad Z(\overline{G}_3) = p^{-1}(F^\times I_3). \quad (17)$$

If Φ is to be the parameter of π , the central identities are

$$\omega_\pi(\mathbf{s}(a^2 I_2)) = \det \Phi(a), \quad (18)$$

$$\omega_\pi(\mathbf{s}(a I_3)) = \gamma_\psi(a) \kappa_\pi(a), \quad \kappa_\pi^2 = \det \Phi. \quad (19)$$

For rank three, the first equality defines an ordinary character κ_π , since the numerator and γ_ψ have the same Hilbert cocycle. Moreover,

$$\mathbf{s}(a I_3)^2 = (a, a) \mathbf{s}(a^2 I_3), \quad \gamma_\psi(a)^2 = (a, a),$$

so $\kappa_\pi(a)^2 = \omega_\pi(\mathbf{s}(a^2 I_3))$. The discrete-series transfer identifies the last expression with $\det \Phi(a)$; parabolic induction preserves the equality by multiplying the block identities. This proves (19) for the construction below.

Lemma 3.1. *A smooth character λ of $F^\times$ is a square of a smooth character if and only if $\lambda(-1) = 1$. When it is a square it has exactly $|A|$ square roots.*

Proof. If $\lambda = \mu^2$, the condition is necessary. Conversely define a character on $F^{\times 2}$ by $a^2 \mapsto \lambda(a)$. It is well-defined and smooth. Extend it across the finite-index inclusion $F^{\times 2} \subset F^\times$; such an extension exists because $\mathbb{C}^\times$ is divisible, and is smooth because its restriction to an open subgroup is smooth. Its value squared at a is $\lambda(a)$. Two extensions differ by a character of the finite group A , which gives the count. $\square$

This lemma explains both the even characters in (15) and the possible choices of κ in (16). For a fixed projected parameter in rank three the square-root choices represent genuinely different central characters, and must not be discarded.

4 Construction from discrete inducing data

We give the construction before the low-rank tables so that no completeness assertion rests just on a list of examples.

4.1 The established input and its precise use

The local correspondence of Flicker–Kazhdan [7], in the corrected covering-Levi formulation of Zou [10, Proposition 3.6], matches essentially square-integrable representations with fixed compatible central character. Its linear image consists of those whose central characters on every block are trivial on μ_2 . Zou’s Proposition 3.17 gives irreducibility of unitary discrete-series induction and uniqueness of its covering-Levi inducing data. The Langlands classification for covers [8], recalled in [4, §11.5], supplies the unique irreducible quotient after ordering real exponents. The metaplectic tensor product and segment results in [9, §§4–7] underlie these statements.

All these inputs apply here over finite extensions of $\mathbb{Q}_2$ as well. In [10], characteristic zero is imposed from §3; the restriction $p \nmid n$ in later Whittaker computations is not a hypothesis of the results just invoked. We do not use those computations.

There is an important logical distinction. The discussion after Proposition 3.20 of [10] does not claim uniqueness of a linear lift for every tempered representation merely from a character identity. Our map is defined by the *unique discrete inducing data*, not by an unproved uniqueness assertion about that broader transfer relation.

4.2 An explicit identification of the discrete building blocks

For an allowed block $B = \rho \boxtimes S_m$ of dimension d , ordinary local Langlands for GL_d [11] gives

$$\Delta(B) = \mathrm{St}_m(\sigma_\rho), \quad \mathrm{rec}(\sigma_\rho) = \rho,$$

with the usual centered segment normalization; an unramified twist in ρ is included in σ_ρ . Its central character is $(\det \rho)^m$, so (5) is exactly the condition for a metaplectic inverse. Thus $\mathrm{MC}^{-1}(\Delta(B))$, with the specified compatible central character, names a uniquely determined genuine essentially discrete representation. This describes the supercuspidal rows as explicitly as the ordinary GL_d correspondence itself, without a false assumption that every supercuspidal comes from an admissible pair for a field extension.

For completeness the transfer can be recognized by its character identity. Let $d_r = \gcd(2, r-1)$ and $d_F(k) = k|k|_F^{-1/2}$; for $k = 1, 2$ this equals $|F^\times/F^{\times k}|^{1/2}$. Let $x \mapsto x^*$ be the conjugacy-compatible power lift of [7, §4] and [10, §3.1]. With Δ_G the square root of the Weyl discriminant, the defining equality for x^2 regular semisimple is

$$\Delta_G(x^2)\Theta_{\tilde{\delta}}(x^*) = \frac{1}{d_F(d_r)|2|_F^{r/2}} \sum_{t^2=x^2} \Delta_G(t)\Theta_{\Delta}(t)\epsilon(x^*/t^*). \quad (20)$$

The same formula holds for covering Levi subgroups, with the ambient central character retained. This specifies the inverse-transfer labels in the tables; existence and uniqueness of these discrete inverse transfers are the cited theorems. We do not pretend to prove the trace-formula theorem anew.

4.3 Forward construction and inverse

Write a genuine irreducible π using its Langlands data. First express each tempered factor by its unique unitary discrete covering-Levi support. Flattening the factors gives unitary essentially discrete blocks $\tilde{\delta}_i$ with real exponents s_i , in decreasing order. Keep the actual representation of the covering Levi and the global genuine central character ω_π ; separate block covers generally do not commute, so their product is the *metaplectic tensor product*.

Transfer these discrete blocks using the covering-Levi correspondence and form

$$\Phi_\pi = \bigoplus_i \text{rec}(\text{MC}(\tilde{\delta}_i)) \nu^{2s_i}. \quad (21)$$

The factor 2 is necessary:

$$\text{MC}(\tilde{\delta} \otimes \nu^s) = \text{MC}(\tilde{\delta}) \otimes \nu^{2s}. \quad (22)$$

Indeed the transfer evaluates the genuine character over a square, and a determinant twist there has value $|\det x|^{2s}$. The same statement with ν^s replaced by any character $\mu \circ \det$ gives the square $\mu^2 \circ \det$. In particular the quadratic ambiguities in choosing separate block representatives disappear in (21); the equivalence criterion for the actual covering-Levi tensor products is [10, Theorem 2.1(4)].

The two outputs are Φ_π in rank two and (Φ_π, κ_π) in rank three, with κ_π from (19). Every indecomposable block is allowed, by the image condition in the discrete transfer.

Conversely let $\Phi \in \mathcal{P}_2^+$ or $(\Phi, \kappa) \in \tilde{\mathcal{P}}_3^+$. Decompose Φ uniquely into indecomposable blocks B_i . Write each as $B_i^u \nu^{t_i}$ with B_i^u having bounded Weil part and $t_i \in \mathbb{R}$. Prescribe the global genuine central character by (18) or (19). For the Levi with these block sizes, apply the discrete metaplectic correspondence to the full essentially discrete tuple $\Delta(B_i)$ and this global character. It gives a unique compatible genuine covering-Levi datum, up to the block choices already absorbed by the metaplectic tensor product. Its i th block has unitary part lifting $\Delta(B_i^u)$ and real twist $\nu^{t_i/2}$. Order the t_i and take the genuine Langlands quotient. Equal exponents can first be grouped using irreducible unitary discrete induction.

The central compatibility condition is exactly $\omega_\pi(\mathbf{s}(a^2 I_r)) = \prod_i \omega_{\Delta_i}(a)$. In rank two it determines ω_π uniquely; in rank three it is equivalent to $\kappa^2 = \det \Phi$. Consequently the inverse exists and has no missing central choice. Uniqueness of discrete support, of the discrete transfer for the actual covering Levi, and of the Langlands quotient proves that these two constructions are inverse. This proves the bijective assertions in Theorem 1.2 for $\eta = 1$.

5 Rank two: the complete matching

Let $I(\lambda_1, \lambda_2)$ be normalized Borel induction of the genuine torus representation (15). The labels are values on *squared* coordinates. In particular labels $\xi \nu^{1/2}, \xi \nu^{-1/2}$ arise from actual rank-one inducing characters with exponents $1/4, -1/4$, not $1/2, -1/2$.

The representation $I(\lambda_1, \lambda_2)$ is irreducible precisely when $\lambda_1/\lambda_2 \neq \nu^{\pm 1}$. At a reducibility point $I(\xi \nu^{1/2}, \xi \nu^{-1/2})$, $\xi \in X^+(F)$, there are two constituents: the essentially discrete one $\delta(\xi)$ and the exceptional Langlands quotient $\theta(\xi)$. Here and below “quotient” uses the standard order of real exponents.

Genuine irreducible representation	Refined parameter Φ_π
$I(\lambda_1, \lambda_2), \lambda_i \in X^+(F), \lambda_1/\lambda_2 \neq \nu^{\pm 1}$	$\lambda_1 \oplus \lambda_2$
$\theta(\xi), \xi \in X^+(F)$	$\xi \nu^{1/2} \oplus \xi \nu^{-1/2}$
$\delta(\xi), \xi \in X^+(F)$	$\xi \boxtimes S_2$
Supercuspidal $\text{MC}^{-1}(\xi \text{St}_2), \xi(-1) = -1$	$\xi \boxtimes S_2$
Supercuspidal $\text{MC}^{-1}(\sigma_\rho), \dim \rho = 2, \rho$ irreducible, ρ $\det \rho(-1) = 1$	

Each unordered pair in the first row is counted once. In the second row the single exceptional quotient accounts for the character-sum parameter at reducibility. The fourth row is essential: genuine supercuspidality does not force the projected parameter to have trivial SL_2 part.

To verify the two S_2 rows, put $\Delta = \xi \text{St}_2$. Its central character is ξ^2 , hence it always admits a discrete inverse. The cuspidality criterion [10, Proposition 3.16] says that this inverse is supercuspidal exactly when the order of $\xi|_{\mu_2}$ equals the segment length 2. Thus it is supercuspidal exactly for $\xi(-1) = -1$, and for $\xi(-1) = 1$ it is the toral discrete constituent $\delta(\xi)$. Irreducible two-dimensional Weil parameters produce the last row by ordinary local Langlands and discrete transfer.

The possible decompositions of a dimension-two parameter are a sum of two characters, an irreducible two-dimensional Weil representation, or $\xi \boxtimes S_2$. Condition (5) gives exactly the displayed restrictions. Hence the table is exhaustive on both sides and realizes the inverse in §4 term by term.

6 Rank three: the complete matching

All rows in the following table include a character κ with $\kappa^2 = \det \Phi$ and have central character $\gamma_\psi \kappa$ on scalar lifts. Let $J_\kappa(\dots)$ denote the irreducible Langlands quotient formed from the displayed covering blocks, using their metaplectic tensor product and that central character. A rank-one block with parameter $\lambda \in X^+(F)$ can be represented by any genuine character $\epsilon \boxtimes \mu$ of the split cover $\overline{G}_1 = \mu_2 \times F^\times$ with $\mu^2 = \lambda$. Different quadratic choices give the same resulting covering-Levi datum once κ is fixed.

In the table, σ_ρ denotes the linear supercuspidal corresponding to an irreducible Weil representation ρ ; $\delta(\chi)$ and the exceptional genuine supercuspidal $C(\chi) = \text{MC}^{-1}(\chi \text{St}_2)$ are as in §5. The letter $C_2(\rho) = \text{MC}^{-1}(\sigma_\rho)$ denotes the other rank-two supercuspidal type. An entry with a rank-one block means the square-root representative just described, not an ordinary tensor product on noncommuting blocks.

Genuine representation	Projected parameter Φ	Conditions
Principal-series Langlands quotient of $\tau_\psi(\lambda_1, \lambda_2, \lambda_3; \kappa)$	$\lambda_1 \oplus \lambda_2 \oplus \lambda_3$	$\lambda_i(-1) = 1$
$J_\kappa(C_2(\rho), \lambda)$	$\rho \oplus \lambda$	$\dim \rho = 2$ irreducible; $\det \rho(-1) = 1, \lambda(-1) = 1$
$J_\kappa(\delta(\chi), \lambda)$	$(\chi \boxtimes S_2) \oplus \lambda$	$\chi(-1) = 1, \lambda(-1) = 1$
$J_\kappa(C(\chi), \lambda)$	$(\chi \boxtimes S_2) \oplus \lambda$	$\chi(-1) = -1, \lambda(-1) = 1$
Genuine supercuspidal $C_{3,\kappa}(\rho) = \text{MC}^{-1}(\sigma_\rho)$	ρ	$\dim \rho = 3$ irreducible; $\det \rho(-1) = 1$
Genuine generalized Steinberg $\delta_{3,\kappa}(\chi) = \text{MC}^{-1}(\chi \text{St}_3)$	$\chi \boxtimes S_3$	$\chi(-1) = 1$

The final row is the essentially discrete constituent associated with the segment of genuine rank-one characters

$$\mu\nu^{-1/2}, \quad \mu, \quad \mu\nu^{1/2}, \quad \mu^2 = \chi, \quad (23)$$

and the prescribed central character. Its square-evaluation labels are $\chi\nu^{-1}, \chi, \chi\nu$. This is consistent with (22). We name its discrete constituent without identifying it with the Langlands quotient of a descending principal series, which is a different constituent.

For completeness, there are exactly five decomposition shapes in dimension three: three characters; an irreducible two-dimensional Weil representation and a character; χS_2 and a character; an irreducible three-dimensional Weil representation; or χS_3 . The S_2 shape splits into

two representation-theoretic rows according to $\chi(-1)$. These exhaust the table. A rank-three genuine supercuspidal cannot lift to χSt_3 : the relevant order on μ_2 is 1 or 2, so it cannot equal the segment length 3. The inverse of χSt_3 is thus the toral discrete representation (23). This also explains why its existence forces $\chi(-1) = 1$.

In the single GL_3 coordinate (8), the refined parameter is

$$\Sigma_\pi = \kappa_\pi^{-1} \Phi_\pi, \quad \Phi_\pi = \Sigma_\pi (\det \Sigma_\pi)^{-1}, \quad \kappa_\pi = (\det \Sigma_\pi)^{-1}. \quad (24)$$

Thus the extra character is not lost under a true isomorphism of dual groups. It is lost if one incorrectly keeps just the first projection $H_3 \rightarrow \text{GL}_3$.

7 From the tables to the Weil-group conjecture

Using the split realizations in (12), the map in the supplied conjecture is now completely specified:

$$\mathcal{L}_1(\pi)(w) = (R_{\Phi_\pi}(w), w), \quad r = 2, \quad (25)$$

$$\mathcal{L}_1(\pi)(w) = ((R_{\Phi_\pi}(w), \kappa_\pi(w)), w), \quad r = 3. \quad (26)$$

It is a homomorphism and a splitting because the projection to W_F is the identity. Continuity follows from the parameter construction. The determinant identity persists under (3), since S_m has determinant one on SL_2 .

7.1 Finite fibers, with sharp bounds

Fix a semisimple Weil representation R . Its possible monodromy operators satisfy

$$R(w)NR(w)^{-1} = \nu(w)N. \quad (27)$$

The possible parameters above it are the orbits of such nilpotent N under the centralizer of R . In dimensions at most three, a nonzero N can link only one-dimensional Weil constituents. Indeed a link between distinct irreducibles of dimension at least two would require total dimension at least four. A link from an irreducible to itself would imply an isomorphism $\rho \simeq \rho\nu$, impossible by determinants. The characters form finite strings under twisting by ν , which has infinite order.

In dimension two, there is at most one link, and it is either zero or nonzero. In dimension three with three distinct characters, at most two successive links occur; each may be zero or nonzero, and diagonal conjugation normalizes every nonzero link to one. This gives at most four orbits. If a character repeats, the only possible link is a map between multiplicity spaces of total dimension at most three; its rank is zero or one, giving at most two orbits. All other decompositions give just $N = 0$.

In rank three fixing the Weil splitting also fixes κ . The centralizer in H_3 acts on N through the full centralizer in GL_3 : for any commuting matrix g , a square root of $\det g$ exists in $\mathbb{C}$. Thus passing to H_3 creates no additional orbits. Since the refined correspondence is injective, these counts prove the bounds in Theorem 1.2.

7.2 The fibers at the reducibility points

For $\xi \in X^+(F)$, the rank-two Weil parameter

$$R = \xi\nu^{1/2} \oplus \xi\nu^{-1/2}$$

has exactly two preimages: $\theta(\xi)$ and $\delta(\xi)$. In their refined parameters, N is respectively zero and a nonzero Jordan block. If $\xi(-1) = -1$, only the nonzero-monodromy parameter is allowed; its preimage is the exceptional genuine supercuspidal $C(\xi)$.

For rank three, fix $\chi \in X^+(F)$, $\kappa^2 = \chi^3$, and $R = \chi\nu \oplus \chi \oplus \chi\nu^{-1}$. The following are the four refined parameters above the same Weil splitting:

$$\begin{aligned} & \chi\nu \oplus \chi \oplus \chi\nu^{-1}, \\ & (\chi\nu^{1/2} \boxtimes S_2) \oplus \chi\nu^{-1}, \\ & \chi\nu \oplus (\chi\nu^{-1/2} \boxtimes S_2), \\ & \chi \boxtimes S_3. \end{aligned} \tag{28}$$

Retain the same κ in every line. The tables give respectively the principal-series quotient, the two different discrete-rank-two-plus-rank-one Langlands quotients, and the generalized Steinberg representation. All four exist and are different. This proves sharpness without relying on a general statement about nilpotent orbits.

7.3 The actual image among Weil splittings

The image can be read without retaining N . In rank two it consists of:

- irreducible R with $\dim R = 2$ and $\det R(-1) = 1$;
- $R = \lambda_1 \oplus \lambda_2$ with both characters even;
- $R = \xi\nu^{1/2} \oplus \xi\nu^{-1/2}$ with $\xi(-1) = -1$.

In rank three the splitting is a pair (R, κ) with $\kappa^2 = \det R$. Its image consists of:

- irreducible R of dimension three with $\det R(-1) = 1$;
- $R = \rho \oplus \lambda$, where ρ is irreducible of dimension two, $\det \rho(-1) = 1$, and $\lambda(-1) = 1$;
- a sum of three even characters;
- $R = \chi\nu^{1/2} \oplus \chi\nu^{-1/2} \oplus \lambda$, with $\chi(-1) = -1$ and $\lambda(-1) = 1$.

In these statements a character is even exactly when it belongs to $X^+(F)$. All determinant constraints follow from the listed conditions. Their smooth square roots exist by Lemma 3.1.

For an explicit omitted parameter choose an odd character α , which exists by extending the nontrivial character of μ_2 . In rank two, $\alpha \oplus \alpha$ is outside the image. In rank three, $(\alpha \oplus \alpha \oplus 1, \alpha)$ is a valid H_3 -valued Weil parameter, but is outside the image: two equal odd characters cannot form the successive pair required in the last bullet. This illustrates why a finite-to-one map must not be silently replaced by a surjection onto the entire target.

8 Transport to every incarnation parameter

For $\eta_i = \eta(e_i - e_{i+1})$, put

$$h_r = 1, \quad h_i = \prod_{j=i}^{r-1} \eta_j^{-1}, \quad \xi(y) = \prod_i h_i^{y_i}. \tag{29}$$

Then $\xi|_{Y^{\text{sc}}} = \eta^{-1}$. The change

$$(y, u) \longmapsto (y, \xi(y)u)$$

is an isomorphism of the bisector extensions carrying the η root data to the trivial root data. By incarnation it gives $f_\eta : \overline{G}_{r,\eta} \simeq \overline{G}_{r,1}$; on torus lifts $y(a)$ its multiplier is $(\xi(y), a)$, and it respects the canonical unipotent splittings. This is the concrete version of [4, §2.9 and §5.3].

For π on $\overline{G}_{r,\eta}$ set $\pi_1 = \pi \circ f_\eta^{-1}$, apply (25) or (26), and transport through the induced isomorphism of L -groups:

$$\mathcal{L}_\eta(\pi) = {}^L f_\eta \circ \mathcal{L}_1(\pi_1), \quad {}^L f_\eta : {}^L \overline{G}_{r,1} \longrightarrow {}^L \overline{G}_{r,\eta}. \quad (30)$$

Both transports are bijections, so the fiber bounds are unchanged.

The remaining effect in the explicit coordinates is small. In rank two $\xi(L)$ consists of squares, so the induced change of the displayed extension coordinates is trivial. In rank three,

$$\xi(y) = \eta_1^{-y_1} \eta_2^{-(y_1+y_2)}, \quad [\xi(y)] = [\eta_1]^{S(y)} \quad (y \in L).$$

Thus the change of the split L -group coordinates is the central quadratic factor

$$w \longmapsto (I_3, (\eta_1, \text{rec}_F(w))) \in H_3.$$

The same factor relates the transported scalar central characters, so (30) retains the appropriate central coordinate. In particular, η contributes neither new representation types nor additional fibers. Fixing (29) makes every transport in the note explicit; no independence-of-arbitrary-choice assertion is needed for the existential conjecture.

9 Normalization checks and use of the supplied volume

The following points prevent common misidentifications in using the tables.

1. The degree is 2, while r is the matrix rank; they must not both be denoted n . The cover used here has $Q(\alpha^\vee) = -1$ and KP parameter $c = 0$.
2. A rank-one character μ transfers to μ^2 . All real inducing exponents therefore double under transfer. This explains the quarter-exponents in the reducible rank-two principal series.
3. The rank-three dual group is the pair model H_3 . Its first projection has kernel μ_2 ; omitting κ merges distinct genuine central characters.
4. The Weil-only assertion is proved using the infinitesimal restriction (3). A stronger refined matching keeps N ; it cannot be replaced by a bijection with Weil splittings alone.
5. The symbol Irr on the left of Conjecture 0.0.3 is interpreted as the genuine set fixed earlier in the supplied note. The refined assertions proved here do not concern nongenuine representations.

For a direct reading path through the second supplied PDF, its printed page number is 18 less than its one-based PDF page number. The relevant locations are:

Purpose	Location in the supplied volume
Historical transfer and its caveats	Gan–Gao–Weissman, §6.4, pp. 18–19 (PDF 36–37)
Weil versus Weil–Deligne parameters	Weissman, §5.2, p. 82 (PDF 100)
Removing η for GL_r	Gan–Gao, §2.9, p. 203 (PDF 221)
The two L -group extensions	Gan–Gao, §5.2–5.3, pp. 213–215 (PDF 231–233)
Covering tori and genuine characters	Gan–Gao, §8.2–8.3, pp. 227–228 (PDF 245–246)
The weak conjecture and reduction	Gan–Gao, §11.1–11.5, pp. 237–239 (PDF 255–257)
KP lattices, dual groups, distinguished character	Gan–Gao, §16.2, pp. 267–268 (PDF 285–286)

The later paper [12, §6.2, Lemma 6.3] also constructs the lift of a projected KP parameter using its genuine central character. The elementary proof in §3, combined with the all-residue-characteristic input of §4, is used here so that no tame restriction from the surrounding wavefront-set discussion is imported. The precise ingredients extending the 2018 volume are [9, Theorem 7.5] and [10, Propositions 3.6, 3.14, 3.16, 3.17].

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